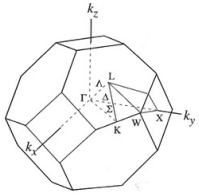
Frequency Response of MOSCAP



Ideal C-V Characteristics

- Accumulation mode:

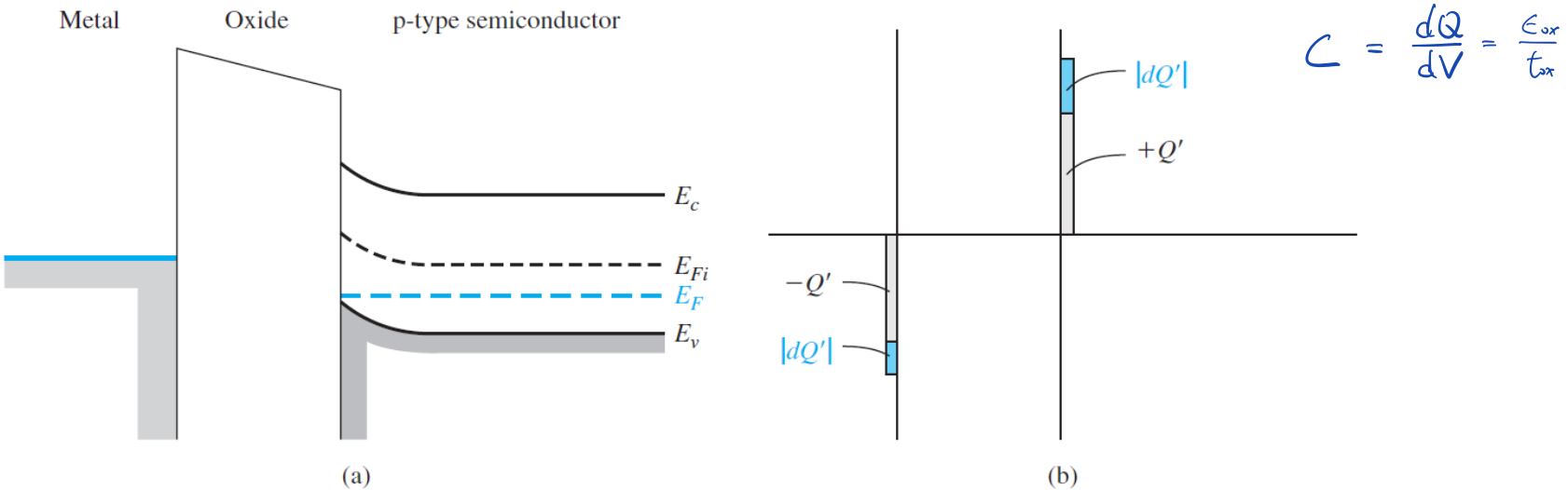
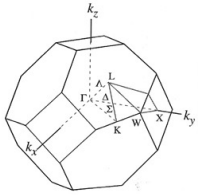


Figure 10.23 | (a) Energy-band diagram through a MOS capacitor for the accumulation mode. (b) Differential charge distribution at accumulation for a differential change in gate voltage.



Accumulation mode:

$$C'(\text{acc}) = C_{\text{ox}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}}}$$

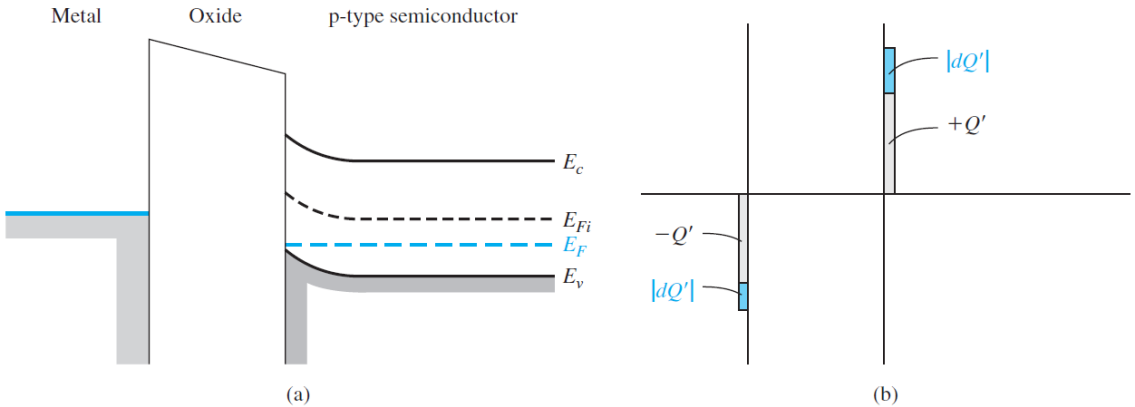
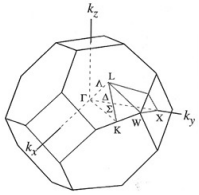


Figure 10.23 | (a) Energy-band diagram through a MOS capacitor for the accumulation mode. (b) Differential charge distribution at accumulation for a differential change in gate voltage.



Depletion Mode

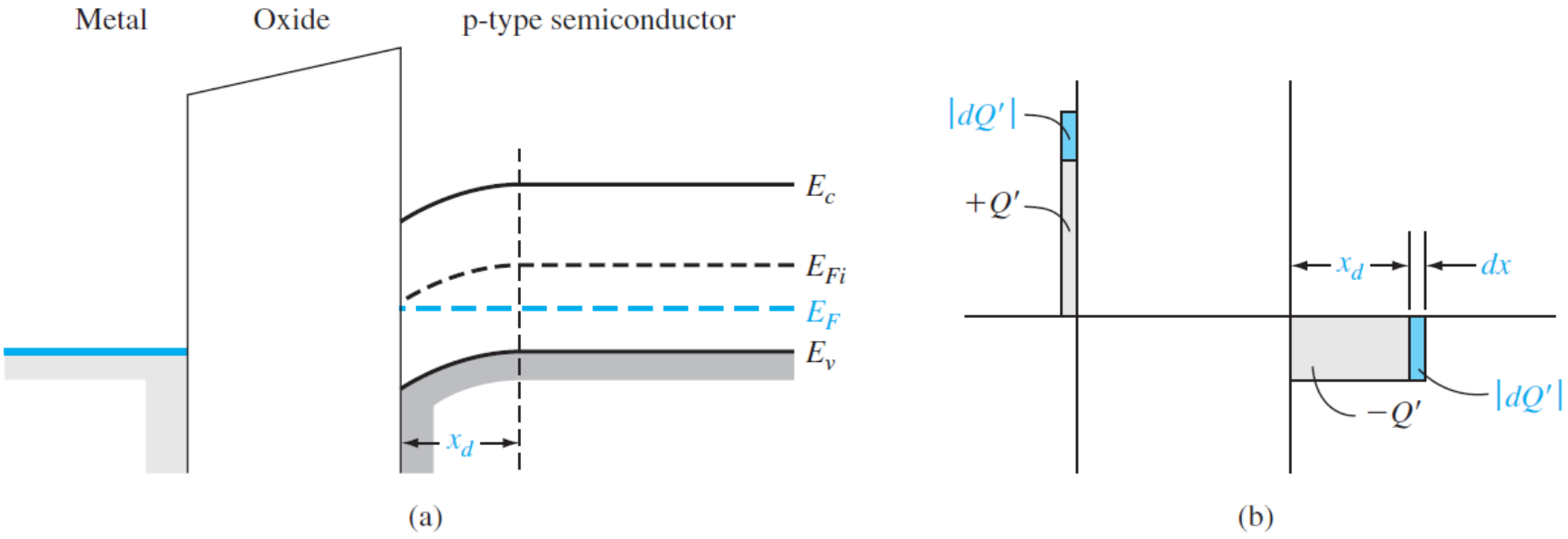
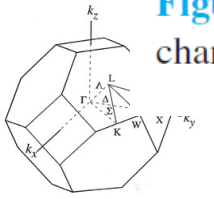
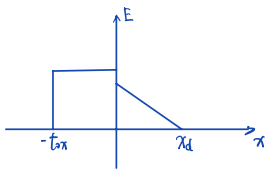


Figure 10.24 | (a) Energy-band diagram through a MOS capacitor for the depletion mode. (b) Differential charge distribution at depletion for a differential change in gate voltage.





$$V_G = t_{ox} \cdot E_{(0-)} + \frac{1}{2} E_{(0^+)} \cdot x_d \qquad Q = eN_A \cdot x_d$$

$$V_G = \frac{eN_A \cdot t_{ox}}{\varepsilon_{ox}} \cdot x_d + \frac{1}{2} \cdot \frac{eN_A}{\varepsilon_s} \cdot x_d^2$$

$$C_{total} = \frac{dQ}{dV_G} = \frac{eN_A \cdot dx_d}{\left(\frac{eN_A t_{ox}}{\varepsilon_{ox}} + \frac{eN_A x_d}{\varepsilon_s}\right) dx_d} = \frac{1}{\frac{t_{ox}}{\varepsilon_{ox}} + \frac{x_d}{\varepsilon_s}}$$

$$\frac{1}{C_{total}} = \frac{t_{ox}}{\varepsilon_{ox}} + \frac{x_d}{\varepsilon_s} = \frac{1}{C_{ox}} + \frac{1}{C_{SD}}$$

Depletion Mode

$$\frac{1}{C'(\text{depl})} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}}$$

$$C'(\text{depl}) = \frac{C_{ox}C'_{SD}}{C_{ox} + C'_{SD}}$$

$$C'(\text{depl}) = \frac{C_{ox}}{1 + \dfrac{C_{ox}}{C'_{SD}}} = \frac{\epsilon_{ox}}{t_{ox} + \left(\dfrac{\epsilon_{ox}}{\epsilon_s}\right) x_d}$$

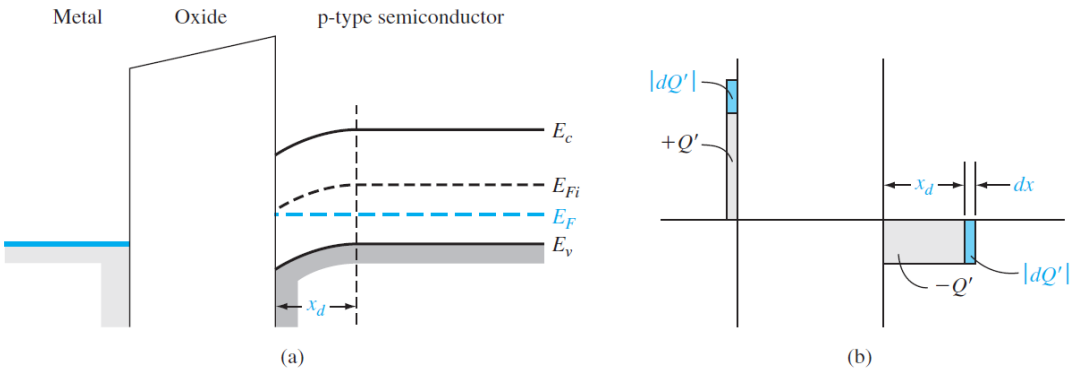
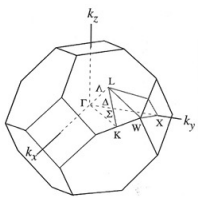


Figure 10.24 | (a) Energy-band diagram through a MOS capacitor for the depletion mode. (b) Differential charge distribution at depletion for a differential change in gate voltage.



Inversion Mode

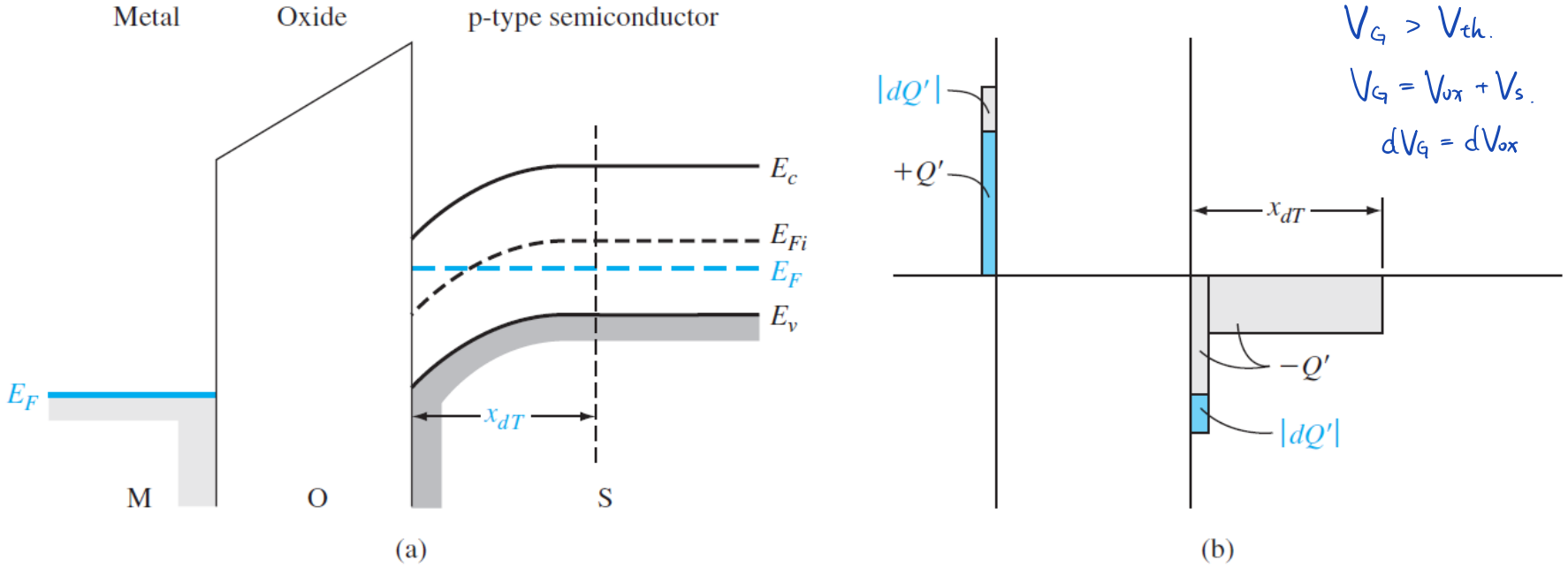
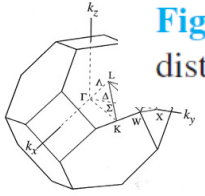


Figure 10.25 | (a) Energy-band diagram through a MOS capacitor for the inversion mode. (b) Differential charge distribution at inversion for a low-frequency differential change in gate voltage.



Inversion Mode

$$C'(\text{inv}) = C_{\text{ox}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}}}$$

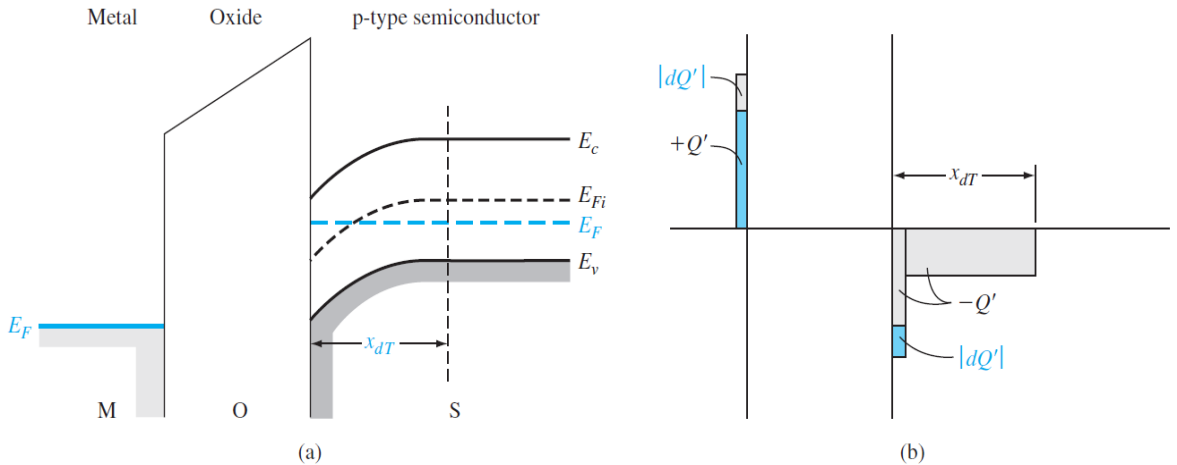
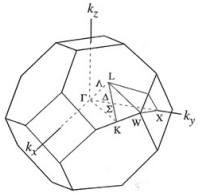


Figure 10.25 | (a) Energy-band diagram through a MOS capacitor for the inversion mode. (b) Differential charge distribution at inversion for a low-frequency differential change in gate voltage.



Ideal Low-Frequency Capacitance

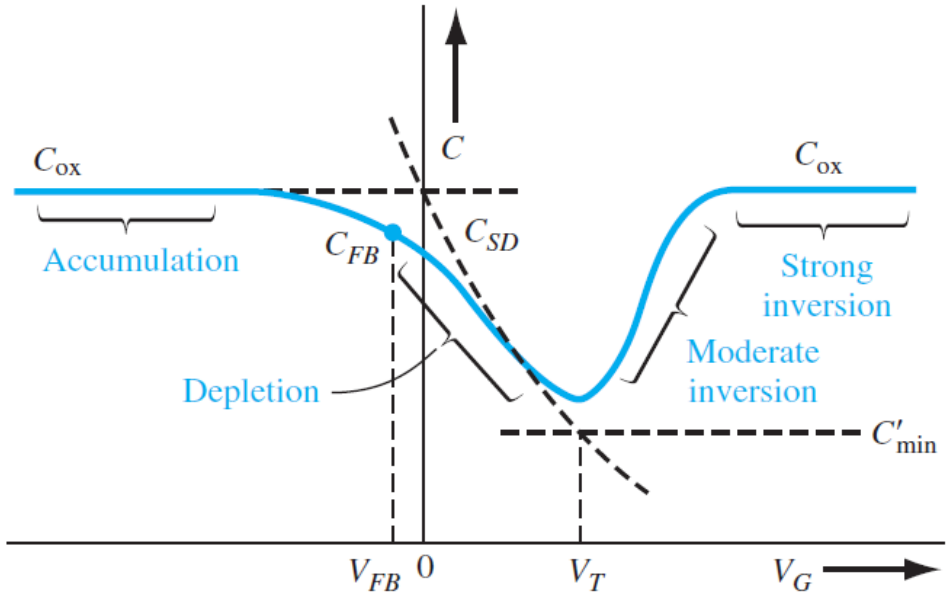
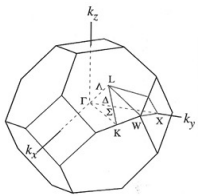


Figure 10.26 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate. Individual capacitance components are also shown.



N-Type Substrate Case

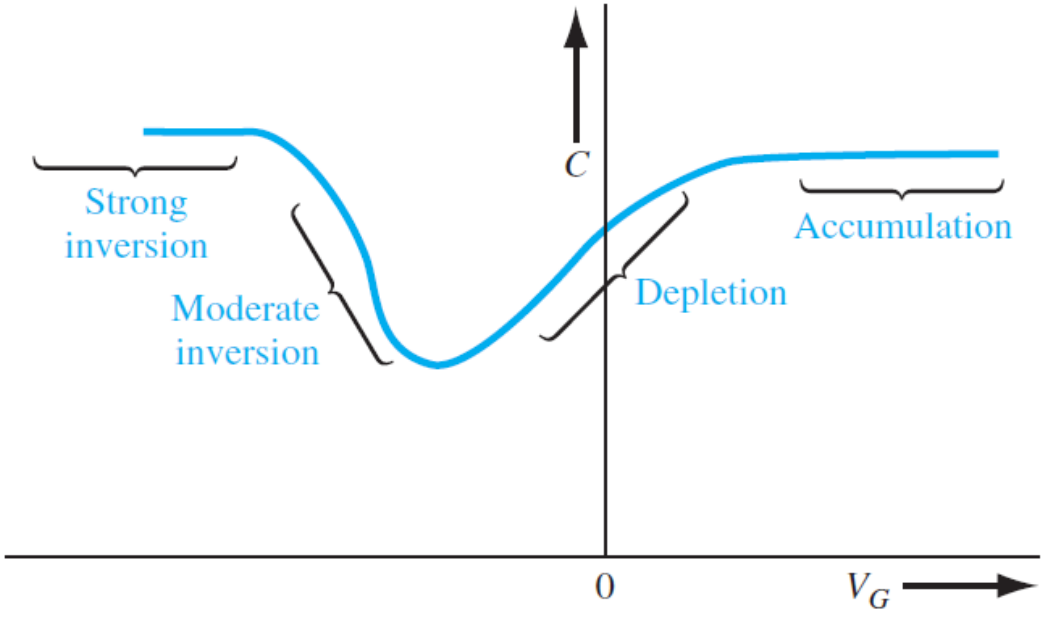
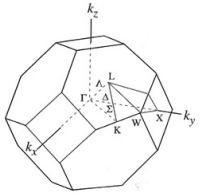


Figure 10.27 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with an n-type substrate.



Frequency Effect

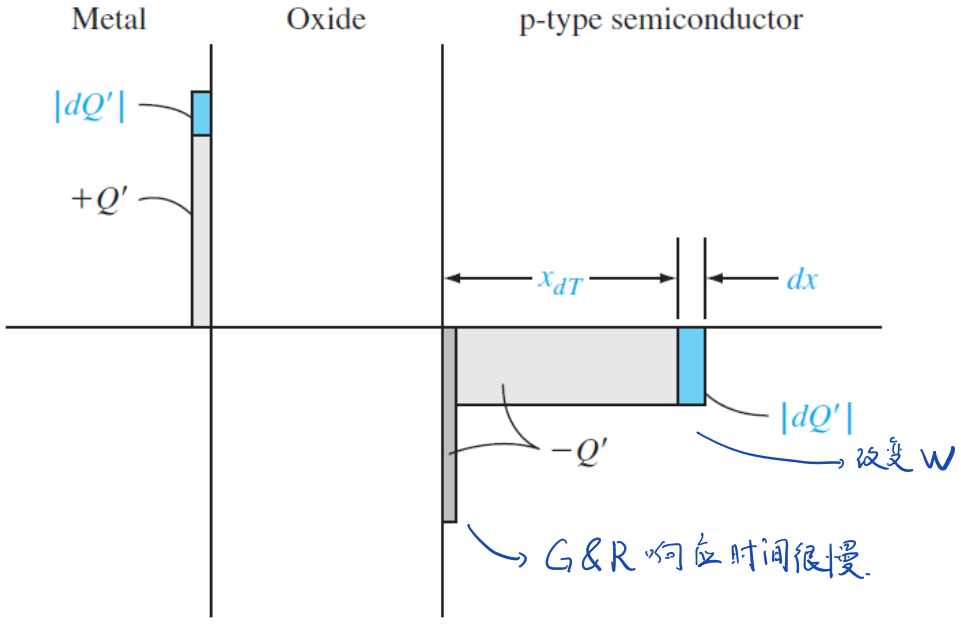
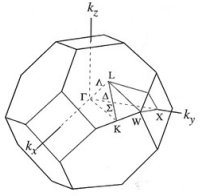
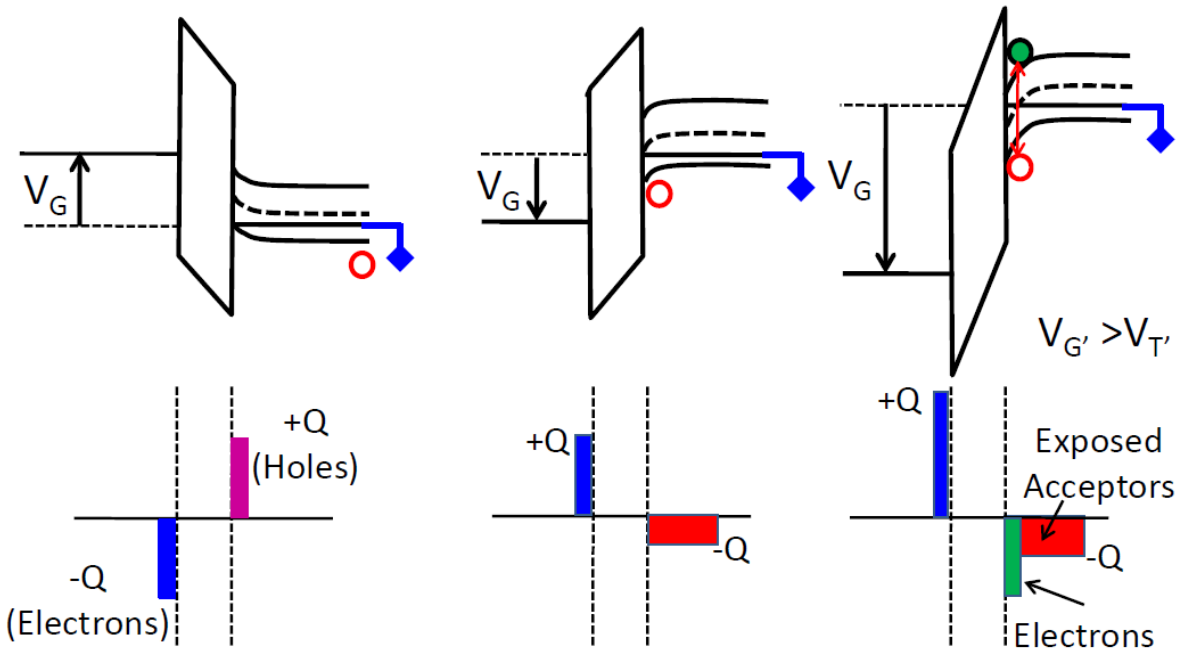


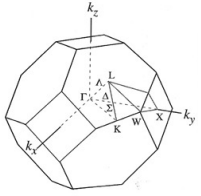
Figure 10.28 | Differential charge distribution at inversion for a high-frequency differential change in gate voltage.



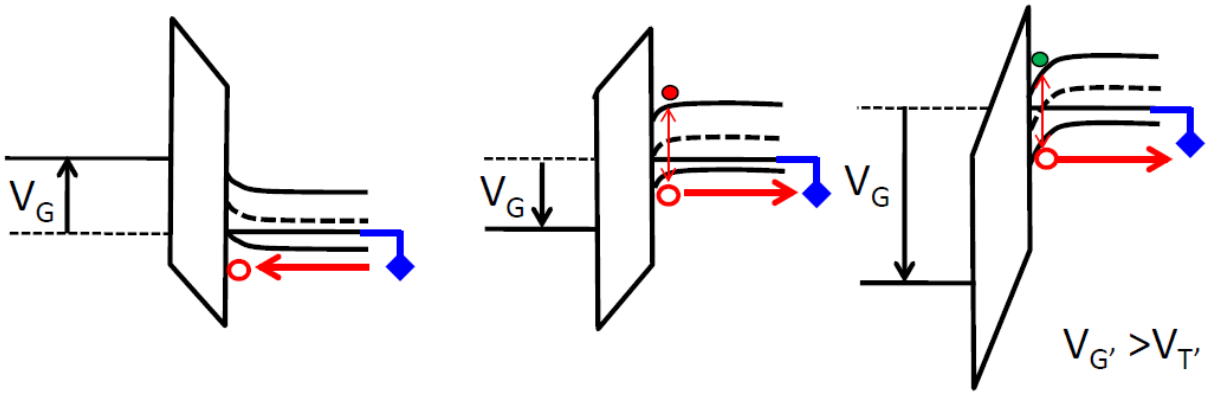
Where Do Charges Come From?



- Integrate charge to find potential.



Response Time

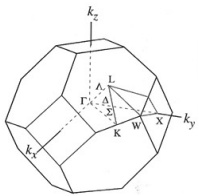


Dielectric Relaxation

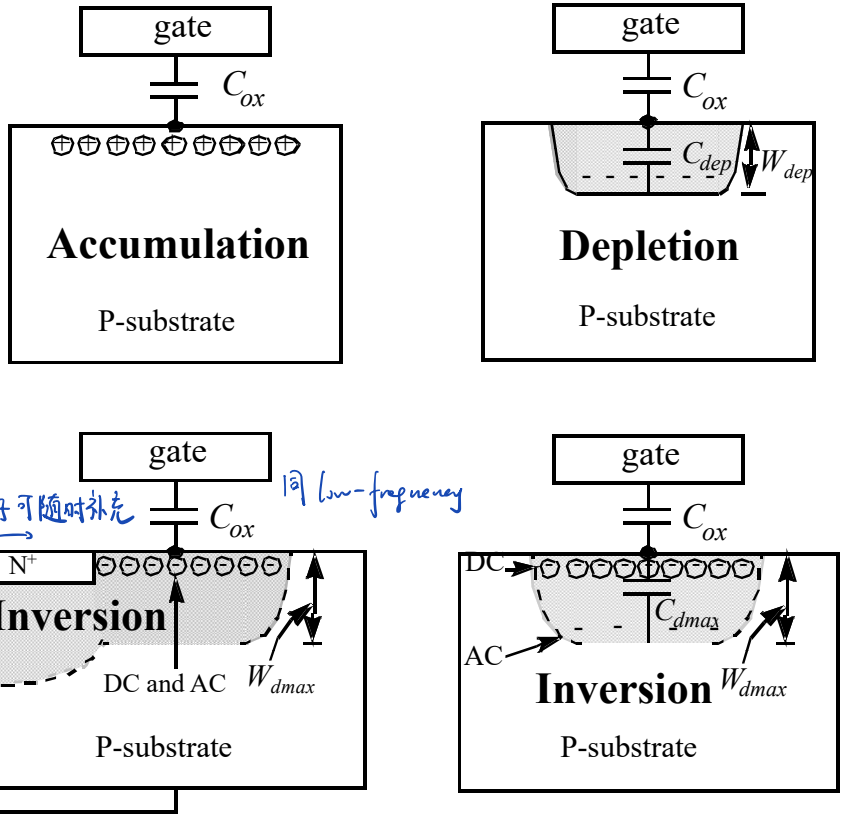
$$\tau = \frac{\sigma}{\kappa_s \varepsilon_0}$$

SRH Recombination-Generation

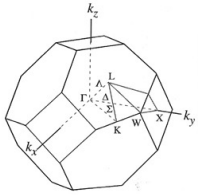
$$R = \frac{np - n_i^2}{\tau_n (p + p_1) + \tau_p (n + n_1)} \rightarrow \frac{-n_i}{\tau_n + \tau_p}$$



Supply of Inversion Charge May be Limited



In each case, C = ?



Frequency Effect

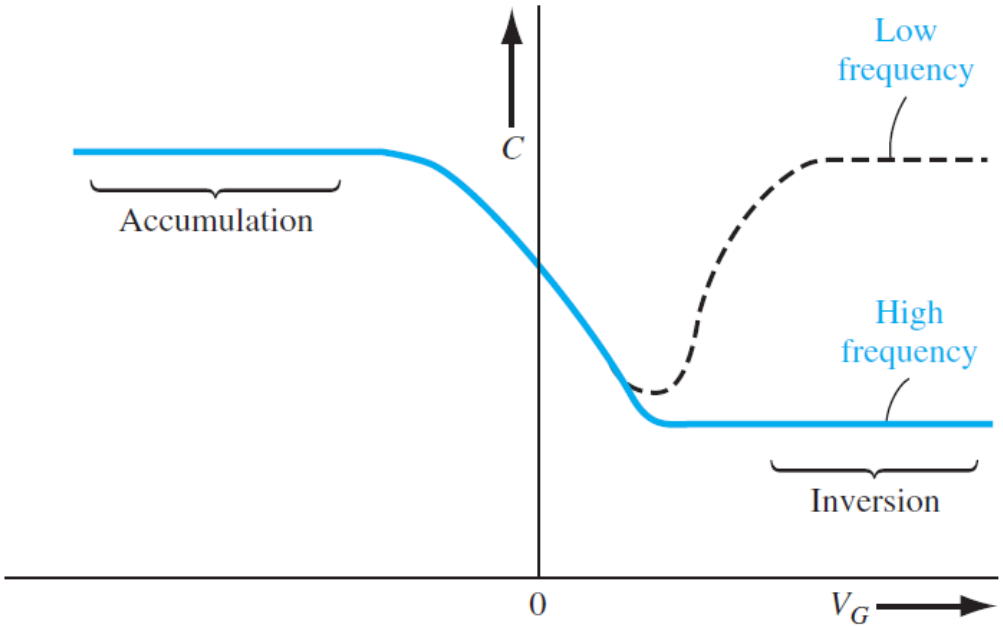
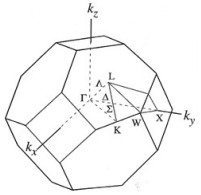
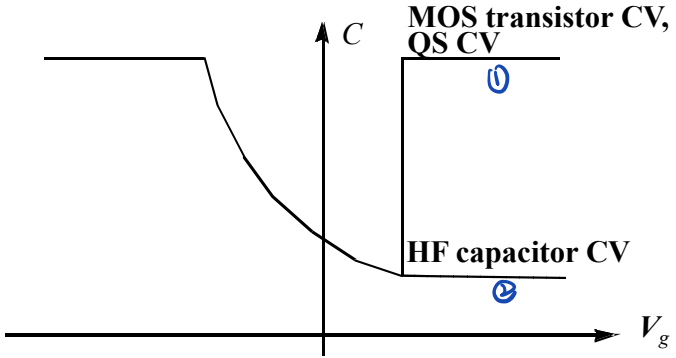


Figure 10.29 | Low-frequency and high-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate.



EXAMPLE : CV of MOS Capacitor and Transistor



Does the QS CV or the HF capacitor CV apply?

(1) MOS transistor, 10kHz. ①

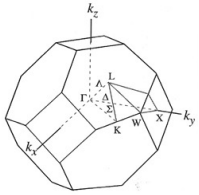
(2) MOS transistor, 100MHz. ①

(3) MOS capacitor, 100MHz. ②

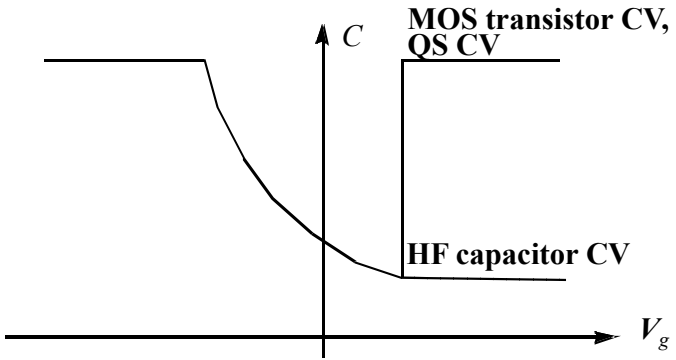
(4) MOS capacitor, 10kHz. ②

(5) MOS capacitor, slow V_g ramp. ①

(6) MOS transistor, slow V_g ramp. ①



EXAMPLE : CV of MOS Capacitor and Transistor



Does the QS CV or the HF capacitor CV apply?

(1) MOS transistor, 10kHz. *(Answer: QS CV).*

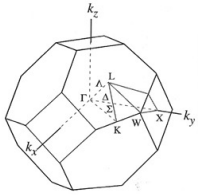
(2) MOS transistor, 100MHz. *(Answer: QS CV).*

(3) MOS capacitor, 100MHz. *(Answer: HF capacitor CV).*

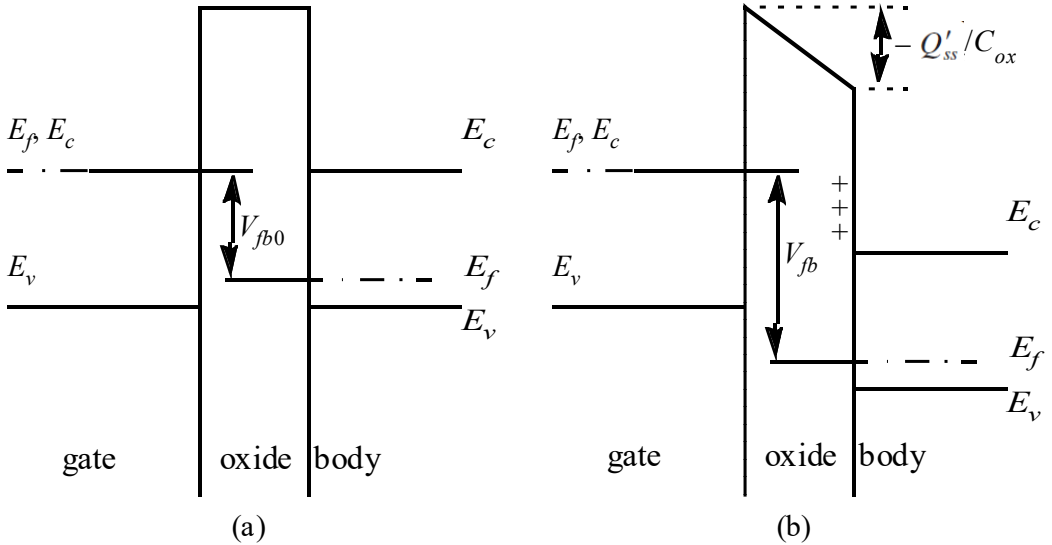
(4) MOS capacitor, 10kHz. *(Answer: HF capacitor CV).*

(5) MOS capacitor, slow V_g ramp. *(Answer: QS CV).*

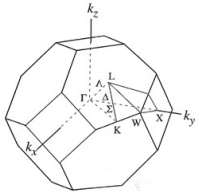
(6) MOS transistor, slow V_g ramp. *(Answer: QS CV).*



Oxide Charge–A Modification to V_{FB} and V_T

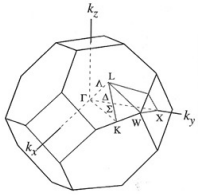


$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$



Oxide Charge–A Modification to V_{FB} and V_T

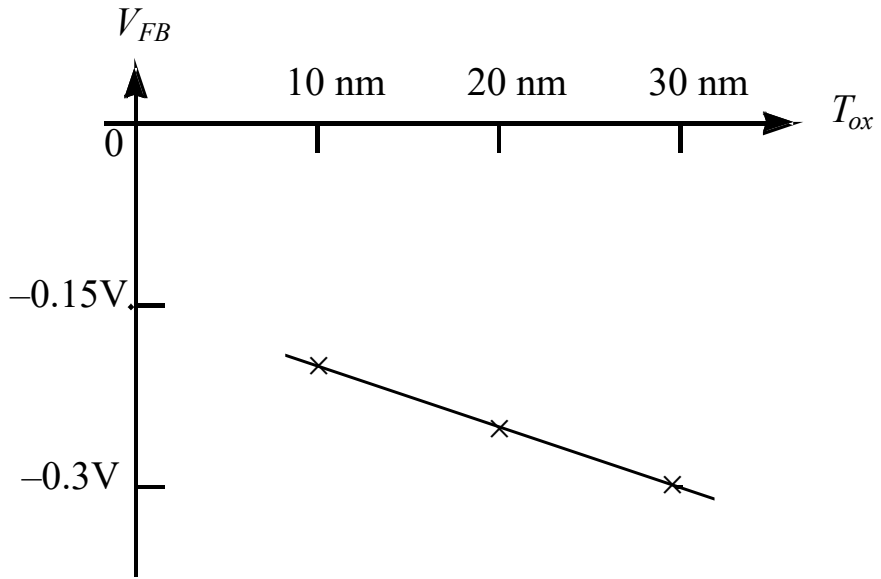
- Types of oxide charge:
  - ➢Fixed oxide charge, Si^+
  - ➢Mobile oxide charge, due to Na^+contamination
  - ➢Interface traps, neutral or charged depending on Vg.
  - ➢Voltage/temperature stress induced charge and traps--a reliability issue



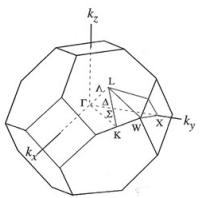
Example

- EXAMPLE: Interpret this measured V_{FB} dependence on oxide thickness. The gate electrode is N+ poly-silicon.

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

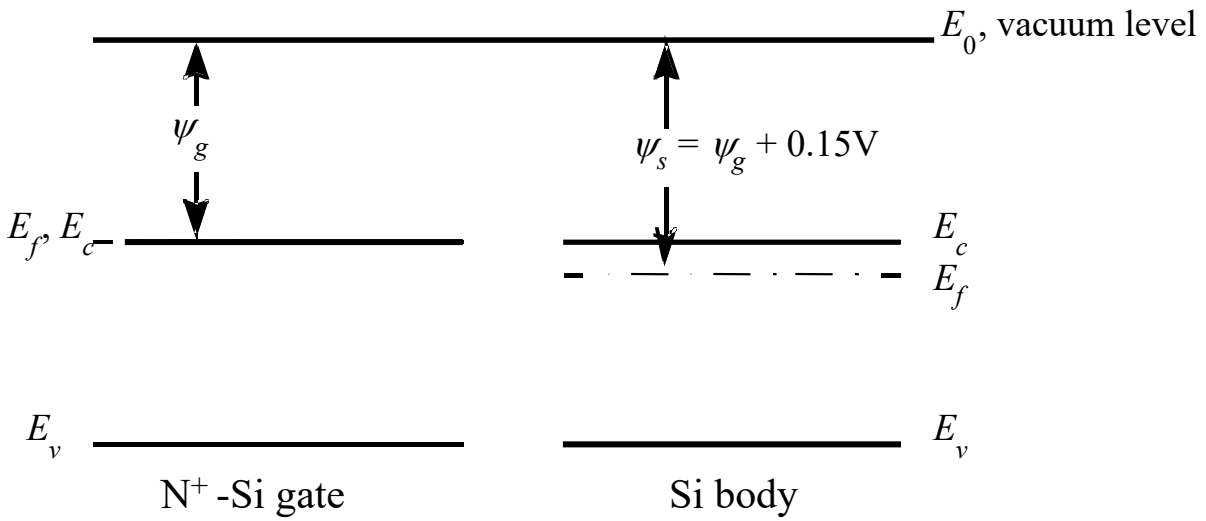


What does it tell us? Body work function? Doping type? Other?



Example

$$\textit{from intercept} \longrightarrow \psi_g - \psi_s = -0.15\ \text{V}$$



$$\textit{N-type substrate,}\quad N_d = n = N_c e^{-0.15\,\text{eV}/kT} \approx 10^{17}\ \text{cm}^{-3}$$

$$\textit{from slope} \longrightarrow Q_{ox} = 1.7\times10^{-8}\ \text{C/cm}^2$$

